KK-L-136b
OCTOBER 8, 1945

SUPERSEDING
Fed. Spec. KK-L-136a
July 24, 1940, and
EMERGENCY ALTERNATE
Fed. Spec. E-KK-L-136a
September 30, 1942

FEDERAL STANDARD STOCK CATALOG

Section IV

(Part 5)

FEDERAL SPECIFICATION

FOR

LEATHER; ARTIFICIAL (UPHOLSTERY)

This specification was approved on the above date by the Director of Procurement for the use of all departments and establishments of the Government, and shall be put into effect not later than December 15, 1945, for use in procurement initiated on or after that date. It may be put into effect, however, at any earlier date after promulgation.

A. APPLICABLE SPECIFICATIONS.

A–1. The following Federal Specifications, of the issue in effect on date of invitation for bids, shall form a part of this specification:

NN-B-621—Boxes; Wood, Nailed and Lock-corner.
UU-P-268—Paper; Kraft, Wrapping.
CCC-T-191 and Supplement thereto—Textiles; General Specifications, Test Methods.

B. TYPES, CLASSES, AND GRADE

B–1. This specification covers the following types and classes of artificial leather for upholstery purposes, as specified in invitation for bids:

- Type I—Pyroxylin, cellulose ether, or cellulose ester coated.
  - Class A—Heavy coated.
  - Class B—Medium coated.
  - Class C—Light coated.
- Type II—Rubber and/or synthetic rubber coated.
- Type III—Synthetic resin coated.

B–2. The grade in each instance shall be that commercially known as "firsts."

C. MATERIAL AND WORKMANSHIP

C–1. *Material.*—

C–1a. *Surface coating.*—

C–1a (1). *Type I.*—The coating shall consist of a pyroxlyin or cellulose ether or ester base suitably compounded.

C–1a (2) *Type II.*—The coating shall consist of a rubber and/or synthetic rubber base suitably compounded.

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C–1a (3). *Type III.*—The coating shall consist of a synthetic resin base suitably compounded.

C–1b. *Fabric base.*—The fabric shall be cotton, and shall be free from sizing except that on the warp yarns necessary for weaving.

C–2. The material shall be free from defects affecting its appearance or which may affect its serviceability.

D. GENERAL REQUIREMENTS

D–1. *Color.*—Artificial leather shall be furnished in color and grain as specified in the invitation to bid.

D–2. *Finish.*—The finish shall be that commercially designated as "medium" or "semibright," unless otherwise specified in invitation to bid.

D–3. *Width.*—The finished material shall be 50 or 54 inches wide unless otherwise specified in the invitation to bid. A minus tolerance of ⅜ inch and any plus tolerance will be allowed.

D–4. *Length of rolls.*—Rolls shall be from 50 to 70 yards in length. Rolls shall not contain more than three pieces, and no piece shall be less than 6 yards in length.

D–5. *Water permeability.*—There shall be no leakage of water through the material under a hydrostatic pressure of 20 inches for a period of ten (10) minutes when tested in accordance with the requirements of paragraph F–2c.

D–6. *Flexibility.*—

D–6a. The artificial leather shall be sufficiently soft and pliable for any upholstery operation.

D–6b. The coating shall not crack through to the fabric base when tested by the method described in paragraph F–2f.

D–7. *Colorfastness to crocking.*—There shall be no appreciable discoloration of the white cloth when tested as described in paragraph F–2d.

D–8. *Resistance to heat.*—

D–8a. *Blocking.*—The coated surfaces shall not be damaged when tested according to paragraph F–2i (1).

D–8b. *Exudation.*—When tested according to paragraph F–2i (2) there shall be no evidence of spue or exudation.

D–9. *Resistance to cold.*—The material shall not crack through to the fabric base at a temperature of 0° F. or at a lower temperature when specified in invitation to bid, when tested in accordance with the method described in paragraph F–2j.

D–10. *Weight tolerance.*—A tolerance in the weight of finished material of plus 2 or minus 1 ounce per square yard will be allowed.

E. DETAIL REQUIREMENTS

E–1. The width, weight, and breaking strength of the finished material shall comply with the requirements as given in Table I.

E–2. *Adhesion of coating film to backing.*—

E–2a. *Type I.*—The pull necessary to separate the coating from the base fabric shall be not less than 3 pounds per inch of width as determined by test covered by paragraph F–2b (1).

E–2b. *Type II.*—The pull necessary to separate the rubber base coating from the base fabric shall be not less than 4 pounds per inch of width as determined by test covered by paragraph F–2b (2).

E–2c. *Type III.*—The pull necessary to separate the coating from the fabric shall be not less than 4 pounds per inch of width as determined by test covered by paragraph F–2b (2).

E–3. *Abrasion resistance. Types II and III.*—The surface coating shall not be abraded down to the fabric backing when subjected to 300 double rubs for type II and 500 double rubs for type III as described in paragraph F–2g.

TABLE I.—*Width, weight, and breaking strength of finished artificial leather*

Type	Class	Width	Weight	Weight of coating (minimum)	Breaking strength, grab method (minimum)	
					Warp	Filling
		Inches	*Oz./sq. yd.*	*Oz./sq. yd.*	*Pounds*	*Pounds*
I	A	50 or 54	19	8	135	125
	B	50 or 54	16	6.5	125	125
	C	50 or 54	11	5	85	45
II	----	50 or 54	19	9	140	120
III	----	50 or 54	19	9	140	120

E–4. *Resistance to aging.*—

E–4a. *Oxygen bomb test. Types II and III.*—The material, after being subjected to the oxygen bomb test as described in paragraph F–2h, shall not be hard and brittle or soft and tacky, and the coating shall not crack through to the fabric base when the specimen is bent coated side out over a ⅛″ diameter mandrel.

E–4b. *Heat test. Type III.*—When fire-resistant material is specified in invitations for bids, the breaking strength of the material after the heat test described in paragraph F–2i (3) shall be not less than 75 percent of that of the material as received.

E–5. *Volatile material. Types I and III.*—The loss in weight of the coated fabric shall not exceed 5 percent when tested by the method described in paragraph F–2k and there shall be no appreciable change in flexibility.

E–6. *Fire-resistance. Type III.*—When fire resistant material is specified in invitations for bids, no specimen, when tested by the method described in paragraph F–2l, shall continue flaming for more than 2 seconds after the burner flame is withdrawn. The average length of char of the 10 specimens tested shall be not more than 2 inches, and the maximum length of char of any one specimen shall be not more than 3 inches.

6a **E–7.** *Mildew resistance. Type III.*—The finished material shall show no visible growth of mildew after being exposed to the mildew infection tests described in Federal Specification CCC–T–191 and Supplement thereto; and the material shall retain a breaking strength of not less than 90 percent of that of unexposed material.

F. METHODS OF SAMPLING, INSPECTION, AND TESTS

F–1. *Sampling.*—A sample, at least one yard long by the full width of the material, shall be taken at random from each delivery of 1,000 yards or fraction thereof unless the lot is in excess of 10,000 yards in which case 10 samples shall be taken, provided, however, one sample shall be taken from each type, class, and color in a delivery.

F–2. *Methods of test.*—

F–2a. *Weight of coating.*—All weighings shall be made on material which is in equilibrium with an atmosphere having a relative humidity of 65 percent and temperature of 70° F. A tolerance of plus or minus 2 percent is permitted in relative humidity and 2° F. in temperature.

F–2a (1). *Type I.*—The weight of the coating shall be determined by removing the coating with acetone, ethyl acetate, or other suitable solvent, drying the cloth, and deducting the weight of the cloth from the weight of the finished material.

F–2a (3). *Type III.*—The weight of coating shall be determined by removing the rubber from the fabric as follows: Soak the back of the fabric with gasoline or other suitable solvent to assist in stripping the coating from the fabric. The fabric shall then be immersed in hot kerosene (approximately 180° Fahrenheit) for 3 to 5 hours to remove all traces of rubber and then rinsed thoroughly in clean gasoline or other suitable solvent and dried. The weight of the fabric will be deducted from the total weight of the finished material.

F–2a (3). *Type III.*—The weight of coating shall be determined by removing the coating from the fabric as follows: Soak the back of the fabric with cyclohexanone or other suitable solvent to assist in stripping the coating from the fabric. Immerse the fabric from which the coating has been removed in hot cyclohexanone (approximately 180° F.) for 3 to 5 hours and then rinse thoroughly in methyl ethyl ketone and dry. The weight of the fabric shall be deducted from the total weight of finished material.

F–2b. Adhesion of coating film to backing.

F–2b (1). *Type I.*—Not less than two specimens shall be taken from each sample with the long dimension in the direction of the warp yarns, and not less than two with the long dimension in the direction of the filling yarns. Each specimen shall measure approximately 4 inches by 9 inches. The coating shall be carefully cut through to the fabric on two parallel lines, 2 inches apart running lengthwise of the piece, but extending so that one cut ends 2 inches from one end of the piece, and the other cut ends 3 inches from the same end. The ends of these lines shall be joined with a diagonal cut through the coating but not the fabric. The edge of the knife shall be carefully worked under the point of the angle thus obtained lifting the coating at the edge of the 2-inch strip for a distance 2 inches from the point. If necessary a small amount of acetone, ethyl acetate or other suitable solvent may be used between the fabric and the coating as an aid in lifting the coating. If a solvent is used, a period of 24 hours should elapse before testing the specimen. The raised end of the coating and the end of the piece shall be clamped in a tensile testing machine equipped with jaws not less than 2 inches wide. The coating shall be pulled from the fabric by the machine which is operated at a speed of separation of the jaws of 12±1 inch per minute. If necessary for satisfactory operation, the coating strip may be reinforced by cementing to it a 2-inch strip of the fabric. The load necessary to strip the 2-inch width of coating from the fabric shall be noted. The results on warp specimens of each sample shall be averaged and divided by two to obtain the load per inch of width necessary to separate the coating from the fabric, warp direction, for the sample. The results for the filling specimens shall be computed in the same way.

F–2b (2). *Types II and III.*—The procedure is the same as that given in paragraph F–2b (1) except gasoline or other suitable solvent type II and cyclohexanone or methyl ethyl ketone for type III shall be used in place of acetone or ethyl acetate.

F–2c. *Water permeability.*—A specimen shall be clamped over a circular opening at least 2 inches in diameter in the bottom of a reservoir. Water shall be introduced into the reservoir to a depth of 20 inches and this depth maintained for 10 minutes. The specimen shall then be examined to determine if leakage has occurred.

F–2d. *Colorfastness to crocking.*—Colorfastness to crocking shall be determined in accordance with the applicable section of F. S. CCC–T–191 in effect on date of invitation for bids. This test shall be conducted on specimens of the material in its original condition and after having been subjected to the light-aging test described in paragraph F–2f.

F–2e. *Breaking strength.*—The breaking strength shall be determined in accordance with the requirements of the applicable section of Federal Specification CCC–T–191.

F–2f. *Flexibility.*—One specimen from each sample, approximately 2½ inches by 3 inches in size, shall be exposed to the radiation from a glass-enclosed carbon-arc lamp, such as the Fade-Ometer, with a current of approximately 13 amperes and 140 volts at the arc. The specimen, coated side toward the lamp, shall be placed 10 inches from the arc and the lamp so ventilated that the temperature of the air at the specimen is maintained at approximately 105° F. during the test. The specimen shall be exposed for 125 hours. The specimen shall be allowed to cool for at least 1 hour and then bent, coated side out, over a ⅛-inch diameter rod. Any cracking of the coating compound through to the fabric base shall be noted.

F–2g. *Abrasion.*—One specimen from each sample, approximately 2 inches by 8 inches in size, shall be cut with the long dimension parallel to the warp yarns and tested for resistance to abrasion, using the Wyzenbeek and Staff abrasion wear tester, operating under the following conditions:

	Type II	Type III
Pressure on specimen, pounds	2	2.
Tension on specimen, pounds	6	6.
Abrader	Silicon carbide cloth, Minnesota mining, 240J.	Silicon carbide cloth, Minnesota mining, 240J.
Speed, double rubs per hour	5,000	5,000.
Temperature of room, ° F	70–90	70–90.
Wear on material and discoloration of abrader to be observed after—	300 double rubs	500 double rubs.

If a specimen fails because of imperfections in the weave of the fabric, disregard that test.

F–2h. *Aging. Types II and III.*—Specimens shall be placed for 8 days in oxygen at a pressure of 300, plus or minus 10, pounds per square inch, at a temperature of 158° ±2° F.

F–2i. *Resistance to heat.*—

F–2i (1). *Blocking.*—A specimen 2 x 4 inches shall be doubled, with the coated surfaces facing each other, and placed between two glass plates weighted with a one pound weight, and heated for 30 minutes at 180° ±2° F. in a thermostatically controlled oven. Five minutes shall be allowed for cooling after removal from the oven. The two surfaces shall then be separated and examined for damage.

F–2i (2). *Exudation.*—A cork mat and a suitable weight are heated to 150° ±2° F. in an electric oven. A specimen approximately 2″ x 4″ is placed on the mat, coated face up, and a single piece of cigarette paper, Riz-La X or equal is placed on the coated surface. The weight is placed on the paper and shall be such as to provide a pressure of 1 ounce per square inch uniformly over the surface of the paper. The temperature of the system is maintained at 150° ±2° F. for 30 minutes. At the end of that time the cigarette paper shall be free from oil stains.

F–2i (3). *Breaking strength.*—Two additional sets of specimens for breaking strength shall be exposed for 7 days in a circulating hot air oven at a temperature of 158° ±2° F. The specimens are then tested for breaking strength in accordance with the applicable requirements of Federal Specification CCC–T–191.

F–2j. *Resistance to cold.*—A specimen after exposure for not less than 2 hours to 0° F. or other temperature as may be specified in the invitation for bids, shall be bent sharply flat upon itself, coated side on the outside and visually examined for cracking.

F–2k. *Volatile material. Types I and III.*—The coated fabric shall be conditioned (see F. S. CCC–T–191) and a 3 x 5 inch specimen cut out and weighed. The specimen shall be placed in a circulating hot air oven and held for 5 hours at 220° ±2° F.; cooled, conditioned, and again weighed. The loss in weight is then calculated and the specimen compared for flexibility with the original sample.

F–2l. *Fire resistance. Type III.*—

F–2l. (1). Five specimens, 2 by 12½ inches, shall be cut with their long dimension in the direction of the warp, and five in the direction of the filling. Each lot of five specimens shall be taken from locations approximately equally spaced throughout the sample, and shall not be taken nearer the selvage than one-tenth the width of the artificial leather.

F–2l (2). The specimens shall be suspended vertically from a clamp covering the upper ½ inch of the length. To protect the specimen from drafts, the apparatus shall be enclosed in a sheet-metal shield 12 inches wide, 12 inches deep, and 30 inches high, open at the top, and provided with a vertical sliding glass front. Sufficient room shall be left at the bottom of the front to allow manipulation of the gas burner used in igniting the specimen.

F–2l (3). The specimen shall be suspended with its lower end ¾ inch above the top of a Bunsen or Tirrill gas burner, having a tube ⅜ inch inside diameter, and with the air supply completely shut off, adjusted to give a luminous flame 1½ inches long. The flame shall be applied vertically near the middle of the width of the lower end of the specimen for 12 seconds, then withdrawn, and the duration of flaming in the specimen noted. The length of char in the specimen shall also be measured after extinction of the glow.

F–2l (4). *Length of char.*—The specimen shall be removed from the clamp and hooks shall be inserted in the specimen, one on each side of the charred area, ¼ inch from the adjacent outside edge and ¼ inch from the lower end of the specimen. It is convenient to punch holes in the cloth for the hooks with an office paper punch. A tearing weight, inclusive of the hook, equal to approximately 10 percent of that required to tear the unburned cloth, shall be attached to one hook.

The weight shall be applied to the specimen gently without impact by raising the other hook. The distance from the end of the specimen which was exposed to the flame to the end of the tear made lengthwise of the specimen through the center of the charred area shall be the length of char.

F-2m. *Mildew resistance. Type III.*—Mildew resistance shall be determined in accordance with Section VIII, paragraphs 1 and 2 of Supplement to Federal Specification CCC-T-191. [see par. E-7]

G. PACKING AND MARKING FOR SHIPMENT

G-1. *Packing.*—Unless otherwise specified, the subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowes rate, to the point of delivery called for in the contract or purchase order.

G-2. *Marking.*—Unless otherwise specified, the material shall be marked as follows:

G-2a. *Rolls.*—The manufacturer's name or trademark, the contract or order number, and the type, class and net yardage shall be plainly marked on each roll.

G-2b. *Package.*—Each package shall be tagged with the following information:

Leather, artificial.
Type and class.
Length of contents ______ yards.
Specification No. ______.
Government contract or order number.
Manufacturer's or contractor's name or trade-mark.

H. REQUIREMENTS APPLICABLE TO INDIVIDUAL DEPARTMENTS

H-1. The following departmental specifications of the issue in effect on date of invitation for bids, and special requirements, shall form a part of this specification, and shall be applicable to purchases made under this specification by the respective departments:

H-2. *Army.*—

H-2a.—*Packaging and Packing.*—The kind of packaging and packing shall be as specified in the invitation for bids and contract or purchase order.

H-2b. *Marking.*—Shipping containers shall be marked in accordance with the requirements of U. S. Army Specification No. 100-2, Standard Specification for Marking Shipments by Contractors, copies of which may be obtained without cost upon application to the procuring agency.

H-3. *Navy.*—

H-3a. *Navy Department specifications.*—

General Specifications for Inspection of Material.
39P16—General Specification for Packaging and Packing for Overseas Shipment:
  Section I—Adhesive, water-resistant, for sealing fiberboard Boxes. (JAN-P-101)
  Section V—Boxes, wood, cleated, plywood. (JAN-P-105)
  Section VI—Boxes, wood, nailed. (JAN-P-106)
  Section VII—Boxes, wood, wire-bound. (JAN-P-107)

Section XXV—Barrier-materials, waterproof, flexible, (JAN–P–125)

Section XXVIII—Tape; water-resistant, gummed. (JAN–P–128)

Copies of the above-referred-to specifications may be obtained without cost upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that naval activities should make application to the Supply Officer in Command, Naval Supply Depot, Bayonne, N. J.

Copies of the Navy Shipment Marking Handbook may be obtained by naval activities and others upon application to the Bureau of Supplies and Accounts.

H–3b. *Resistance to salt water.*—The coating shall not be visibly affected by immersion for 24 hours in a saturated aqueous salt solution ($NaCl$), rinsing in water, and allowing to dry. The artificial leather shall not give color to the test solution.

H–3c. *Resistance to gasoline and to oil.*—After exposure to oil for 1 hour or to gasoline for 1 minute, there shall be no indication of the permeation of the fluid. In determining these properties, an 8 x 8-inch specimen shall be placed on a wood frame. The inside dimensions of the wood frame shall be 6 x 6 x 1 inch. The artificial leather shall be inserted with the coated side up. The sample shall be forced into the frame by a wooden block, 5¾ x 5¾ x ¾ inch, with rounded corners to form a basin of uniform depth. The edges of the sample shall be tacked to the frame, and the block removed. The specified fluid shall be poured into the basin as rapidly as possible to a depth of ½ inch. After the fluid has been in the basin for the specified time, the bottom of the specimen forming the basin shall feel dry, and there shall be no indication that the fluid has permeated the artificial leather.

H–3d. *Packing.*—

H–3d (1). *For domestic shipment.*—Unless otherwise specified, the leather, in widths and lengths specified in paragraphs D–3 and D–4, respectively, shall be wound on standard fiberboard cores and wrapped in not less than 2 thicknesses of waterproof material conforming to the requirements of Navy Department Specification 39P16, section XXV (JAN–P–125), for interior wraps. The ends of each roll shall be protected by headers of the same, or equal, material securely glued or taped to the body of the outer wrapping with water-resistant adhesive or water-resistant tape conforming to the requirements of Navy Department Specification 39P16, sections I and XXVIII (JAN–P–101 and JAN–P–128), respectively. The seams and closures shall have strength and water-resistance equal to that of the body material.

H–3d (2). *For overseas shipment.*—Unless otherwise specified, individual rolls of the leather, packed as described in paragraph H–3d (1) shall be overpacked in a cleated plywood box, nailed wood box or wirebound box conforming to the requirements of Navy Department Specification 39P16, sections V, VI, and VII (JAN–P–105, JAN–P–106 and JAN–P–107), respectively. The boxes shall be designed to fit the rolls in a compact manner and shall be strapped in accordance with the requirements of the appendix of the applicable box specification.

H–3e. *Marking.*—

H-3e (1). *Rolls* (for domestic shipment).—Unless otherwise specified, rolls shall be plainly marked with the following information completed:

MATERIAL ______________________________
SIZE ______________________________
TYPE ______________________________
CLASS ______________________________
STOCK NO ______________________________
COLOR ______________________________
CONTRACT NO ______________________________
CONTRACTOR ______________________________
MANUFACTURER ______________________________
WEIGHT ______________________________

H-3e (2). *Shipping containers* (for overseas shipment).—Unless otherwise specified, shipping containers shall be plainly marked in black with the following information completed:

MATERIAL ______________________________
STOCK NO ______________________________
SIZE ______________________________
TYPE ______________________________
CLASS ______________________________
COLOR ______________________________
CONTRACT NO ______________________________
CONTRACTOR ______________________________
CUBE ______________________________
GROSS WEIGHT ______________________________

In addition to the foregoing, marking shall conform to the requirements of the latest issue of the Navy Shipment Marking Handbook.

I. NOTES

I-1. The principal uses of the different classes of type I are as follows:

Classes A and B are intended primarily for ordinary uses of upholstery such as tubular metal furniture, auditorium and theater seats, and home and office furniture of spring type construction, where fire resistance is not a factor.

Class C is intended primarily for slip seats, flat seats with little or no padding, pasted-on constructions, headlining, wall lining, kick pads, and other uses where the material is desired for appearance but receives little wear and where fire resistance is not a factor.

I-2. The principal uses of type II are as follows: This type is intended primarily for all ordinary uses of upholstery in deep spring, semideep spring or sponge rubber seats, such as tubular metal furniture, auditorium and theater seats, automobile seats, and home and office furniture, where fire resistance is not a factor.

I-3. The principal uses of type III are as follows: This type is intended for general uses of upholstery in deep spring, semideep spring, or sponge rubber seats, such as tubular metal furniture, auditorium and theater seats, automobile seats, and home and office furniture, where fire resistance is or is not a factor, and where extremely high resistance to severe flexing and abrasion, and good resistance to oil, grease, most solvents, dilute acids, salt water, and low temperatures are required.

2a I-4. Purchasers should exercise any desired options offered herein and specify (1) the type, class, color, and grain desired; (2) the cold test temperature, if one lower than 0° F. is needed and (3) that type III material shall be fire resistant when that qualification is desired.

I–5. It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

I–6. Federal Specifications do not include all types, classes, grades, sizes, etc., of the commodities indicated by the titles of the specifications, or which are commercially available, but are intended to cover the types, etc., which are suitable for Federal Government requirements.

I–7. An Index of Federal Specifications, and also a Supplement-Index of Emergency Alternate Federal Specifications, may be purchased as noted in the paragraph below, price to be obtained from the Superintendent of Documents.

I–8. Copies of this specification and of NN–B–621, UU–P–268, and CCC–T–191 and Supplement thereto may be obtained upon application, accompanied by money order, coupon, or cash to the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Price 5 cents each; except CCC–T–191—Supplement, 15 cents.

Notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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